

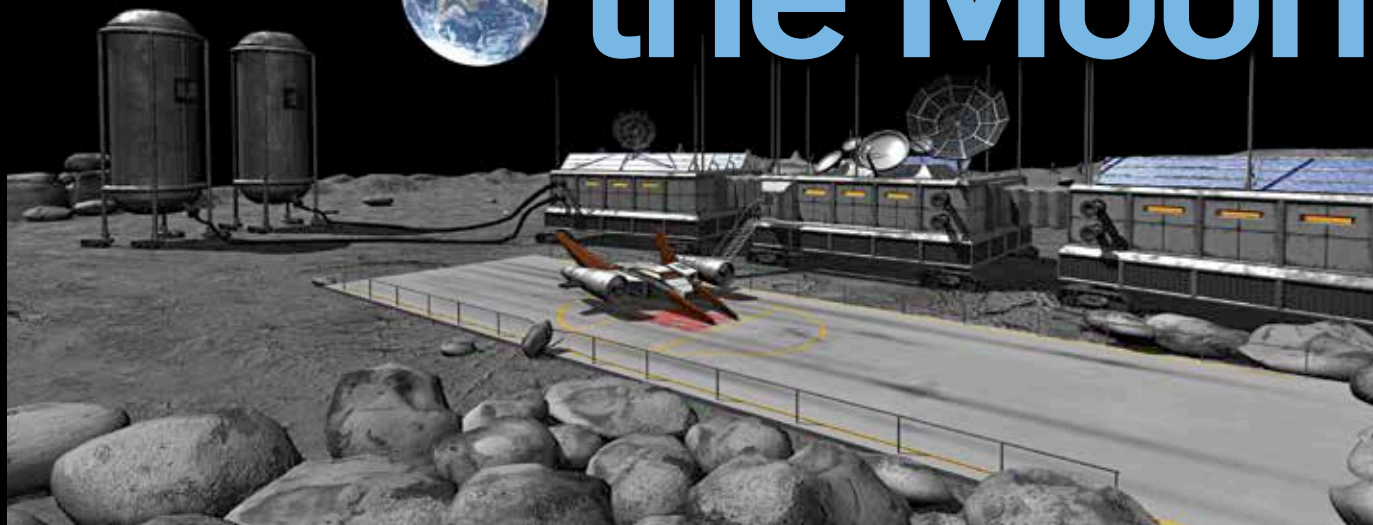
Insight

Human athletes are a cut above the rest, so much that you could call them machines.

**F1 Health Check**

How the health of the car and drivers is monitored in the world of F1.
<http://digit.in/F1health>

The Dark Side of the Moon



The global superpowers are vying for vast deposits of unclaimed energy resources on the moon. The winner will solve the energy problem for the entire world.

Shaan Repswal
readersletters@digit.in

The year 2014 was pretty great for the global space economy, and this has been going for the last two decades. Comprising of ground services, satellite manufacturing, communications, TV, and remote sensing among other things, the global space economy grew by nine percent in 2014. In case you're wondering, that kind of a growth rate is pretty high for any industry. It's a matter of revolutionary proportions if this growth rate can be sustained for a few decades. That's exactly what's been going for the past few decades, and there is no reason for the trend to change.

This economy was valued at a total of \$330 billion worldwide in 2014. That's around three times the size of the global gaming industry. As nations other than the US and Russia invest in their space programmes, the non-US government

space spending has outpaced that of the US government. Total commercial space activities make up to 76 percent of the aforementioned statistic. That means the industry is sustainable and economically sound. The global space industry is extending its reach beyond the sub-orbital business. It is no longer exclusively about bringing 4G internet, Google Maps, Skype and HD Porn to common folk any more. United States, Europe, China, Russia, India and Japan have already declared their wish to mine the moon for precious minerals and set up permanent bases there.

Japan's space Agency called Japan Aerospace Exploration Agency, aka JAXA, has already launched its Hayabusa-2 probe to mine asteroids. If that sounds far fetched, just wait till you hear about what Russia has been up to. Sometime between 2005 and 2006, Nikolai Sevastyanov, the owner of Energia Space Corporation, unveiled plans to build a permanent base on the moon by 2020 to start mining the planet for Helium

3. Then there's the Chinese, who plan on accomplishing this same feat in the next 50 years with the Chang'e program.

Unclaimed Bounty

What's on the moon anyway? Rare Earth Materials, which include fifteen lanthanides from the periodic table plus scandium and yttrium. These elements are mostly used in modern electronics in trace quantities, but considering that there are billions of electronic devices out there these elements are being used up copiously. Rare Earth Materials are, well, rare and the moon's got plenty of that just waiting to be mined.

The most important resource is however the Helium-3 isotope. Known as Tritium, it's primary application would be as fuel for fusion reactors of the future. In fact, scientists at ISRO plan on building a He-3 fusion powered engine for a future space faring vehicle. Tritium is hardly available on the Earth, because the Earth's magnetic field shields us from the highly charged solar winds that carry Tritium. But the upper layers of the moon's regolith has been collecting Tritium over the past billion years. By current estimates there's between one and five million tons of Tritium on the moon. The Earth has just fifteen tons in comparison.